

UC Berkeley  
Master of Climate Solutions

*A program of Rausser College of Natural Resources*

# The MCS Playbook

# Learning objectives

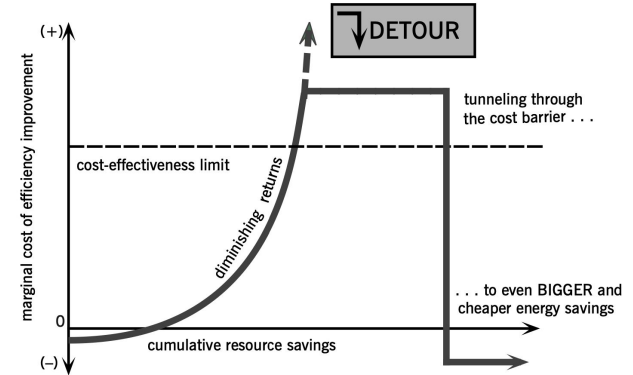
1. Review high level climate solutions strategies
2. Introduce our standard analysis "playbook" for climate solutions technologies
3. Develop a list of "levers" we can pull to address climate solutions costs.
4. Interpret a climate solutions scoping plan in the context of the MCS playbook

Reading for this unit:

- CARB's 2022 Scoping Plan Executive Summary
- Koomey and Monroe, Ch 2 (recommended)

# Climate solutions technology mindset (review):

- Take a systems-level perspective; thinking on the margin won't work
- Ensure solutions align with a just transition



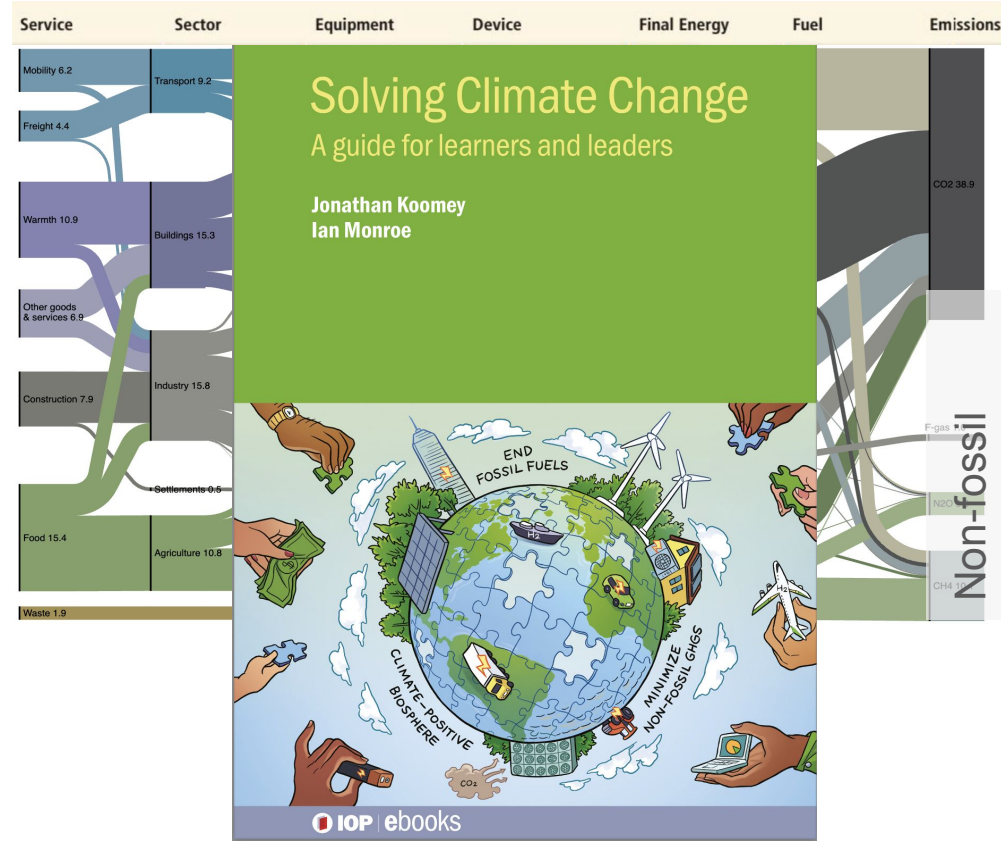
# Climate solutions mindset (from Koomey and Monroe)

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1. Electrify what you can
2. Decarbonize supply (electricity, low carbon fuels)
3. Increase efficiency
4. Remove carbon
5. Minimize non-fossil warming

To support this: Configure regulatory, financial, and media systems to enable



# The MCS Technology Analysis Playbook™

## Step 1. Techno-economic analysis:

- a. Compute the ***technical potential*** of the climate solution in terms of avoided CO<sub>2</sub> emissions
- b. Determine the ***implied cost per ton of CO<sub>2</sub>*** avoided by the climate solution
- c. Enumerate ***caveats*** to the technoeconomic analysis

## Step 2. Evaluate socio-political drivers and barriers:

- a. Stakeholder analysis
- b. Evaluate who could pay the cost, and how

## Step 3. In sum: What is the potential for speed and scale?

- a. What are the most important issues to address for the solution to work quickly and at scale?
- b. What is your assessment of the potential for the solution to deliver gigaton-scale emissions reductions?

# Step 1a: Technical potential

1. Define the *incumbent technology* and its market area
2. **Compute the CO2 emissions** for the incumbent technology in the market area
3. **Quantify the service** delivered by the incumbent technology
  - a. E.g. how much heating in buildings, vehicle miles traveled, metric tons of steel, electricity generated...
4. Compute the CO2 emissions for the climate solution to deliver the **same service** in the **same market area**.

This is where the  
fun modeling is...

# 1a, Technical potential: Example questions

These are pure energy accounting questions. Examples:

- **Demand side:** How much energy does the carbon emitting technology consume? How much zero carbon energy do we need for its substitute?
- **Supply side:** How big is the zero carbon energy resource, how does it compare to what we need?

How big are the **efficiency** penalties? Can they be overcome?

Examples:

- Solar needs (inefficient) energy storage; may need to be curtailed
- Rankine cycle in a nuclear generator is only ~40% efficient.



## Step 1b: Implied cost per ton

$$\text{Cost per metric ton} = \frac{\text{Climate solution cost} - \text{status quo cost}}{\text{Status quo CO}_2 - \text{Climate solution CO}_2}$$

The challenge? Determine what units to use for costs and emissions

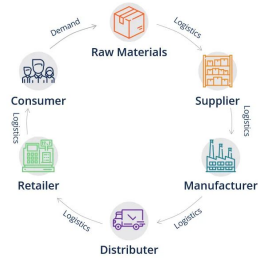
Option 1: Costs and emissions for total market area

Option 2: Costs and emissions ***per unit of stuff***

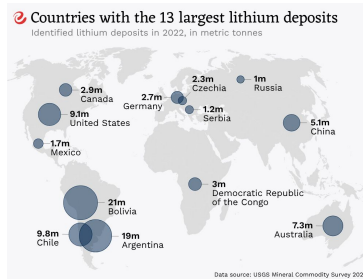
"Stuff" could be kWh, buildings, pounds of steel, airline flight miles, etc.

# Step 1c: What are the techno-economic caveats?

Example Q1: Is the supply chain able to deliver inexpensively?



Example Q2: Do we have the materials?



Example Q3: Is the solution ready for production?



Example Q4: Are there systemic issues that must also be addressed?



## Step 2a: Socio-political drivers and barriers

- Who are the major stakeholders?
- What do stakeholders stand to gain or lose from the solution?
- How does the solution enable (or disable) a just transition?
- What power do the stakeholders have?
- What stakeholder coalitions are possible?

## Step 2b: Who pays and how?

- When we have an implied cost of carbon, how will society pay for it, and how will that impact stakeholders?

Two ways to think about this:

- *Private* costs to the consumers and businesses that purchase and use the technology solution.
- Total *social* cost to deploy the technology solution.

Why would social cost and private costs differ?

- Governments could spend money to support solutions (we'll talk about that in this class)
- Societal externalities (we won't talk about that in this class)

## Group activity: Ways to reduce the private cost of carbon abatement



(Results [here.](#))

# Duncan's summary: ways to offset climate solutions costs

|                          |                                                                                                  |
|--------------------------|--------------------------------------------------------------------------------------------------|
| <b>Reduce capex</b>      | Investment Tax Credits (ITCs), Grants, Accelerated Depreciation.                                 |
| <b>Reduce opex</b>       | Production Tax Credits (PTCs), Feed-in Tariffs (FITs), Renewable Energy Certificates (RECs).     |
| <b>Price carbon</b>      | Carbon Taxes, Emissions Trading Schemes (Cap-and-Trade).                                         |
| <b>Standards</b>         | Renewable Portfolio Standards (RPS), EV Sales Mandates, Energy Efficiency Standards.             |
| <b>Subsidize R&amp;D</b> | Government-funded National Labs, University Grants, Innovation Prizes.                           |
| <b>De-risking</b>        | Loan Guarantees, Direct Green Public Infrastructure Spending, Advance Market Commitments (AMCs). |

## Step 3: In sum, evaluate potential for speed and scale

1. What are the warts to a solution...what needs to improve?
2. What is the size of the total addressable market? (How much carbon can the solution save?)
3. Does the solution have “voltage,” i.e. the power to beat the status quo?  
Beware of confirmation bias as you evaluate the data to answer this question.  
(Cribbed from John List's *Speed and Scale* book)
4. Are people really willing to pay? Be skeptical. What are the realistic opportunities to reduce private costs?

# Class discussion: California ARB

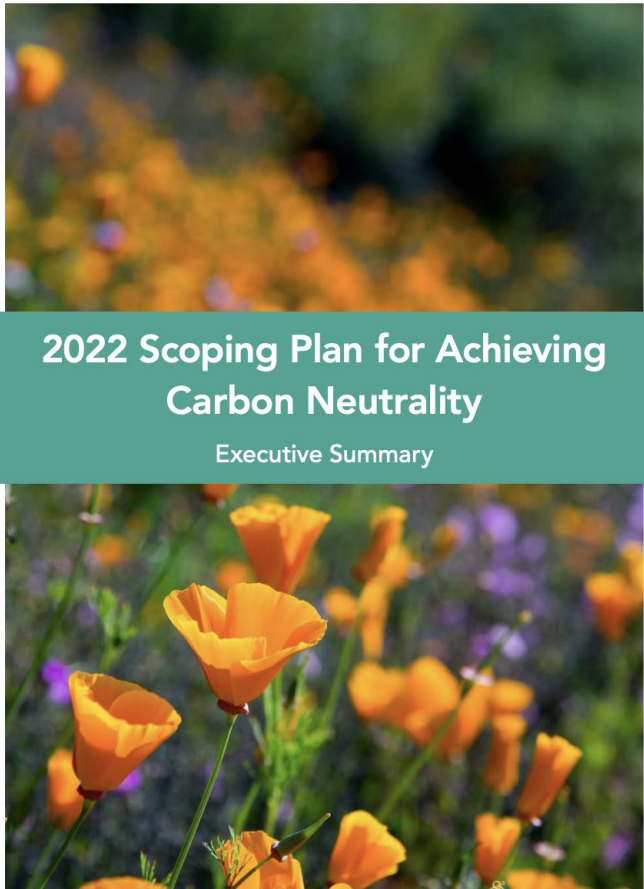
How much and in what ways do they discuss:

- Technical potential
- Economic potential
- Caveats, like materials availability and tech.readiness?

How do they evaluate social, political and business opportunities?

- Stakeholders;
- Their interests, power, and role in a just transition

How do they address potential for speed and scale?



## 2022 Scoping Plan for Achieving Carbon Neutrality

Executive Summary

Quiz, and break